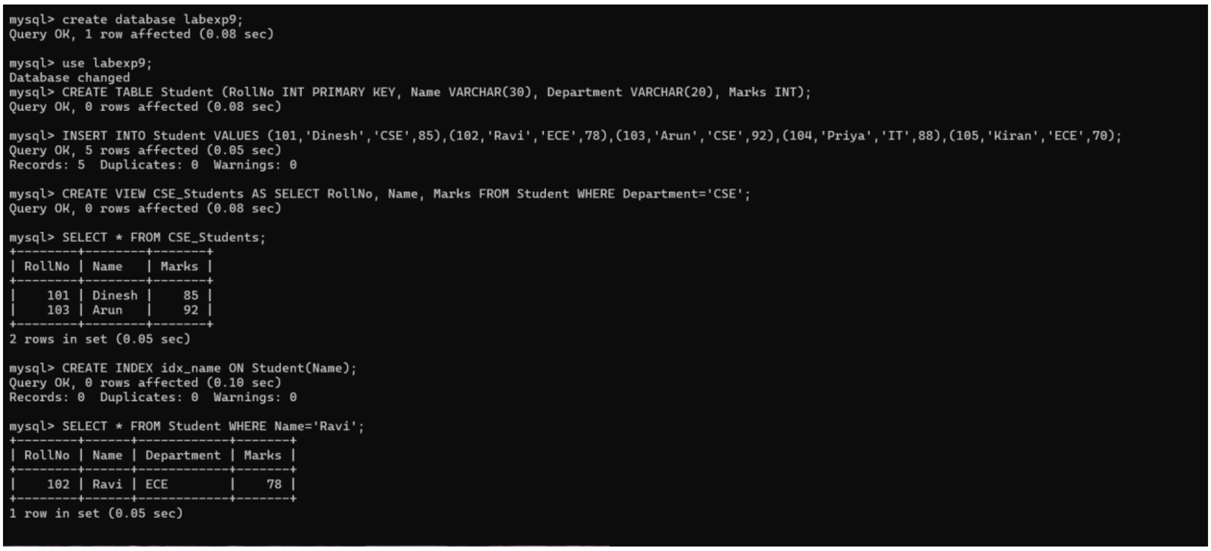
SIMATS ENGINEERING

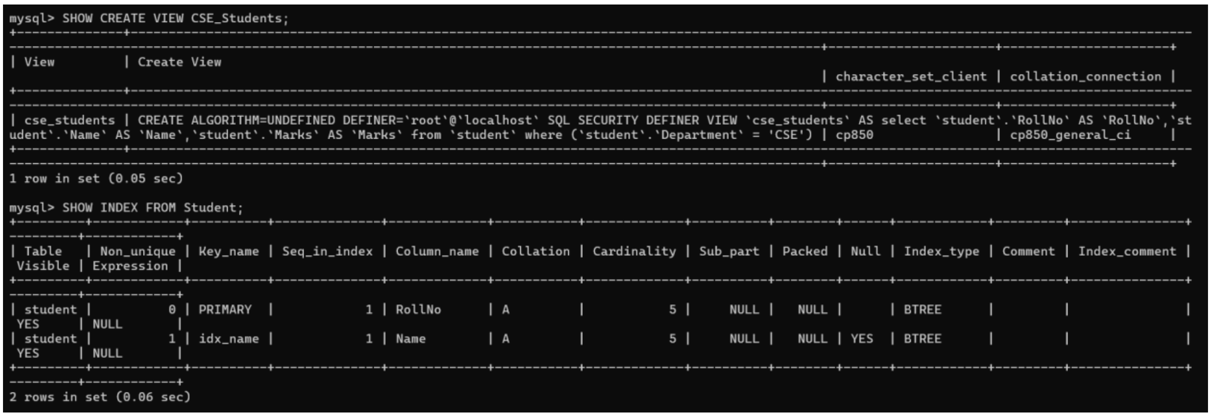
LAB EXPERIMENTS

COURSE NAME: **DATABASE MANAGEMENT SYSTEMS**
COURSE CODE: **CSA0515**

EXP 9 : Database with VIEW and INDEX



```
mysql> create database labexp9;
Query OK, 1 row affected (0.08 sec)

mysql> use labexp9;
Database changed
mysql> CREATE TABLE Student (RollNo INT PRIMARY KEY, Name VARCHAR(30), Department VARCHAR(20), Marks INT);
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Student VALUES (101,'Dinesh','CSE',85),(102,'Ravi','ECE',78),(103,'Arun','CSE',92),(104,'Priya','IT',88),(105,'Kiran','ECE',70);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE VIEW CSE_Students AS SELECT RollNo, Name, Marks FROM Student WHERE Department='CSE';
Query OK, 0 rows affected (0.08 sec)

mysql> SELECT * FROM CSE_Students;
+--------+--------+-------+
| RollNo | Name   | Marks |
+--------+--------+-------+
|    101 | Dinesh |    85 |
|    103 | Arun   |    92 |
+--------+--------+-------+
2 rows in set (0.05 sec)

mysql> CREATE INDEX idx_name ON Student(Name);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student WHERE Name='Ravi';
+--------+------+------------+-------+
| RollNo | Name | Department | Marks |
+--------+------+------------+-------+
|    102 | Ravi | ECE        |    78 |
+--------+------+------------+-------+
1 row in set (0.05 sec)
```



```
mysql> SHOW CREATE VIEW CSE_Students;
+--------------+---------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------+----------------------+----------------------+
| View         | Create View
                                                                                                              | character_set_client | collation_connection |
+--------------+---------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------+----------------------+----------------------+
| cse_students | CREATE ALGORITHM=UNDEFINED DEFINER=`root`@`localhost` SQL SECURITY DEFINER VIEW `cse_students` AS select `student`.`RollNo` AS `RollNo`,`st
udent`.`Name` AS `Name`,`student`.`Marks` AS `Marks` from `student` where (`student`.`Department` = 'CSE') | cp850                | cp850_general_ci     |
+--------------+---------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------+----------------------+----------------------+
1 row in set (0.05 sec)

mysql> SHOW INDEX FROM Student;
+---------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
---------+------------+
| Table   | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
 Visible | Expression |
+---------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
---------+------------+
| student |          0 | PRIMARY  |            1 | RollNo      | A         |           5 |     NULL |   NULL |      | BTREE      |         |               |
 YES     | NULL       |
| student |          1 | idx_name |            1 | Name        | A         |           5 |     NULL |   NULL | YES  | BTREE      |         |               |
 YES     | NULL       |
+---------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
---------+------------+
2 rows in set (0.06 sec)
```